

Appendix: Expert commentary on How to Pull the Right Lever

Class Divide is a community campaign group, not a group of statisticians. We asked some of the country's leading education researchers to review Professor Dennett's report. Their responses are reproduced here in full and unedited, with their permission.

Professor Stephen Gorard, Durham University

The report by Adam Dennett makes some good points which are uncontested, because they are already established. For example, the level of socio-economic segregation between schools in Brighton and Hove is not remarkable, and is known to be worse elsewhere (almost anywhere rural with a lower population density and/or weaker public transport). This is not any argument for not addressing the local segregation, but is perhaps a reminder to other authorities to act in the ways that Brighton and Hove has tried to do for decades. My work has shown a strong link between this segregation and the size of the poverty attainment gap across time and place. Therefore, it is no surprise to note that the attainment gap in Brighton and Hove is also better (smaller) than in many other areas of England. Again, this is not an argument to accept it as it is. It is also well-known that the attainment gap has no single cause. It is rather odd for Dennett to suggest that anyone thinks otherwise. Poverty linked to early health, nutrition, and stresses must be a key factor that leads to both differences in attainment at KS4, and to earlier school outcomes at KS2, KS1 and before.

Dennett treats these prior attainment scores as causal, and so creates an unwitting proxy for early family poverty. Absence from school is a factor, but only one affecting a minority of children. In his account Dennett has confused "coefficients" with "effect" sizes. In a regression model a coefficient (such as for absence) can be large, suggesting that it is important for the minority it affects, but the overall effect size (R or R squared) can be negligible or even zero meaning that it is not important in the overall model. Again, saying this is not argument for not trying to reduce absence as well.

Dennett's model has several problems. The data is reported as being for all schools in England. Therefore, they have neither been randomly selected nor allocated randomly to groups. In the absence of randomisation, the use of significance tests, p-values and ** in tables is mathematically meaningless. They cannot be computed, even though software will try to do so if ordered to erroneously. Basing any conclusions on these misleading figures instead of actual differences, is perhaps worse than focussing on coefficients rather than increases in R. Given that non-randomised figures have no standard error and need no generalisation to a population (the figures are already for the population) the complexity for readers of multi-level modelling is needless. The report says that several variables are not linearly related to the outcomes, but the regression model that was used only assessed linear relationships. Therefore, a low effect size does not mean that two variables are unrelated, only that they are not linearly related.

In assessing a proposed causal model in education, the order of events matters. A cause cannot come after its purported effect. This is why family structure, parental education, ill health, special needs, and above all poverty and the early clustering of poverty at school, would need to be assessed before prior attainment. But pragmatically, it is almost as important to consider the malleability of any determinant of school success. Yes, family poverty in England or disparities in parental education could be addressed but not easily, not within education, and not in the short-term. This is why de-clustering schools by poverty has been described as an "easy win". Changes to admissions arrangements (criteria), use of bussing or lotteries, and Pupil Premium

funding, have all shown that school intakes can be immediately desegregated to some extent. No one loses from this fairer school system (although some parents in Brighton and Hove may feel concerned about their children having to attend with a refreshed intake). So, it will be interesting to see how school intakes change, and whether any change in segregation does, in fact, lead to a lower than expected poverty attainment gap.

Professor Diane Reay, University of Cambridge

Professor Dennett's report separates out a whole range of variables - absence, prior attainment, social segregation, and disadvantage - as if they are independent of each other when they are dynamically interconnected. Poverty leads to non-attendance which then results in low attainment. If you don't address the disadvantage which is actually driving the absences in schools like BACA and Longhill, attendance gains are likely to be minimal and temporary. The current catchment system in Brighton and Hove concentrates privileged parents around a small number of popular schools. Professor Dennett reframes the debate as absence versus disadvantage. But this is not a technical question to be settled by better data but a political and ethical one about how places should be allocated fairly, and who schools should serve.